Observations on Model Results and Variance Across Runs

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1. Bug Fix in pretrain_model()

After running the pipeline for the first time, the fine-tuned results for Legal-BERT and Legal-SBERT were identical to each other, which was unexpected. After investigating, I found an issue in pretrain_model(): load_base_model() was called inside it without any model_name argument, meaning it always fell back to the default model in CONFIG — that being MPNet. In practice, MPNet was being pretrained and fine-tuned three times, with each run overwriting the same output directory.

The fix was to remove the load_base_model() call from pretrain_model() entirely, since the model is already loaded in compare_models() before pretrain_model() is called. I introduced a model input parameter to pass the already-loaded model directly, and a model_name parameter so each model's output is saved to a separate directory. The corrected call in compare_models() is:

trained_model = pretrain_model(baseline_model, cases_sample, model_path)

After applying this fix and re-running, all three models produced different results from each other, confirming the fix worked as intended.

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2. Results After Fix (Run 1)

After the fix, Legal-SBERT performed best after fine-tuning, which seems reasonable given that the legal cases are in German and its base model XLM-RoBERTa was pre-trained on a multilingual corpus.

The results still differed from those in the original project PDF, which is expected — the test set is small enough that even minor differences in predictions shift all metrics noticeably.

Run 0 — MPNet (winner) – results shown in the original ‘LLM Project Course IV.pdf’:

Metric	Baseline	Fine-Tuned	Improvement (%)
Recall@5	0.0075	0.0307	309
Precision@5	0.0667	0.2727	309
F1@5	0.0135	0.0553	309
MRR	0.0687	0.2652	286
MAP	0.0687	0.2467	259

Run 1 — Legal-SBERT (winner):

```
======================================================================
**PERFORMANCE IMPROVEMENT SUMMARY: legal-sbert**
======================================================================
| Metric      |   Baseline |   Fine-Tuned |   Improvement (%) |
|:------------|-----------:|-------------:|------------------:|
| Recall@5    |     0.0082 |       0.0243 |          195.8333 |
| Precision@5 |     0.0727 |       0.2152 |          195.8333 |
| F1@5        |     0.0147 |       0.0436 |          195.8333 |
| MRR         |     0.0922 |       0.1896 |          105.7534 |
| MAP         |     0.0896 |       0.1808 |          101.6901 |
```

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3. Variance Across Runs

I ran the full pipeline two more times to check consistency. The results fluctuated noticeably between runs:

- **Run 1:** Legal-SBERT won (F1@5 = 0.0436)
- **Run 2:** MPNet won (F1@5 = 0.0436)
- **Run 3:** MPNet won (F1@5 = 0.0528)

Run 2 — MPNet (winner):

```
======================================================================
**PERFORMANCE IMPROVEMENT SUMMARY: mpnet**
======================================================================
| Metric      |   Baseline |   Fine-Tuned |   Improvement (%) |
|:------------|-----------:|-------------:|------------------:|
| Recall@5    |     0.0075 |       0.0243 |          222.7273 |
| Precision@5 |     0.0667 |       0.2152 |          222.7273 |
| F1@5        |     0.0135 |       0.0436 |          222.7273 |
| MRR         |     0.0687 |       0.1884 |          174.2647 |
| MAP         |     0.0687 |       0.1821 |          165.0735 |
```

Run 3 — MPNet (winner):

```
======================================================================
**PERFORMANCE IMPROVEMENT SUMMARY: mpnet**
======================================================================
| Metric      |   Baseline |   Fine-Tuned |   Improvement (%) |
|:------------|-----------:|-------------:|------------------:|
| Recall@5    |     0.0075 |       0.0294 |          290.9091 |
| Precision@5 |     0.0667 |       0.2606 |          290.9091 |
| F1@5        |     0.0135 |       0.0528 |          290.9091 |
| MRR         |     0.0687 |       0.2172 |          216.1765 |
| MAP         |     0.0687 |       0.2068 |          201.1029 |
```

Why the variance?

Two reasons:

1. No random seed. The dataloader has shuffle=True during both pretraining and fine-tuning, with no random seed set anywhere. This means batch ordering changes every run, which leads to meaningfully different final model weights.

2. Dataset too small. With only 66 test cases, the models don't have enough data to learn reliably — results depend heavily on whether a given test case happens to be similar to something seen during training. The fact that the same data is used for both fine-tuning and evaluation makes this even more pronounced, as also noted in the original project report.

Summary

Despite the variance, the overall performance level is consistent across all runs. The F1@5 score stays roughly in the range of **0.043–0.055**, which matches what the original report reported. The winning model changes between runs, but no model consistently dominates.

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4. Inference Test: English vs. German Query

As a small additional test, I ran the inference cell twice with the same legal scenario — once in English and once in German — to check whether query language affects retrieval quality. The test was performed using the fine-tuned Legal-SBERT model, which was the winning model of the first run.

The queries used were:

- *"Employer didn't pay the salary to the employee."*
- *"Der Arbeitgeber hat dem Arbeitnehmer kein Lohn ausbezahlt."*

Language	Similarity score range of top 5 results	Top result
English	0.63 – 0.66	§612a BGB
German	0.76 – 0.79	§611a BGB

The German query produced noticeably higher similarity scores and returned a more relevant top result. §611a BGB defines the employment contract and explicitly states the employer's obligation to pay the agreed remuneration (*"Der Arbeitgeber ist zur Zahlung der vereinbarten Vergütung verpflichtet"*), making it directly applicable to a salary dispute. By contrast, §612a BGB addresses the prohibition of employer retaliation, which is less relevant to the described scenario.

This result is consistent with the system's design: the fine-tuning data and the entire statute corpus are in German, and Legal-SBERT's base model XLM-RoBERTa was pre-trained on a multilingual corpus with strong German language representations. It is worth noting, however, that given the very limited size of the training dataset, the model's performance remains sensitive to whether a given query resembles cases it has already encountered during training. The results should therefore be interpreted

cautiously, as this single test case is not sufficient to draw generalizable conclusions about cross-lingual retrieval quality.